

Part 1 – Data Analytics: Understanding Traffic Patterns

Learning Outcomes Addressed:

- Use SQL to analyse real-world datasets and identify meaningful trends, patterns and changes over time or across relevant categories.
- Apply descriptive statistics and correlation analysis to summarise data and examine relationships between variables.
- Apply probability and conditional probability concepts to assess the likelihood of events and relationships between relevant variables.
- Transform, clean and prepare data using Power Query to support effective analysis and visualisation.
- Develop interactive Power BI dashboards that communicate key findings and data-driven insights clearly to non-technical stakeholders.
- Interpret analytical results and translate them into actionable, evidence-based insights to support decision-making in a real-world context.

Tasks

Task 1: SQL-Based Traffic Analysis

Using the Metro Interstate Traffic Volume dataset:

1.1 Load the dataset

- Load the comma-delimited traffic CSV file into **SQLite**.
- Verify that the data has been loaded correctly.

The screenshot displays the ExtendsClass SQLite browser online interface. The top navigation bar includes links for Database file, Queries, Save & Share, Export, and Run. The main area is divided into two sections. The top section shows the 'Your database' sidebar with a 'Table' named 'traffic' and a 'View' button. The central query editor contains the SQL statement:

```
1 SELECT COUNT(*) AS row_count
2 FROM traffic;
3
```

 The bottom section shows the 'SQL Documentation' sidebar with a list of SQL commands: SELECT, UPDATE, DELETE, INSERT, and CREATE TABLE. The central query editor contains the SQL statement:

```
1 SELECT *
2 FROM traffic
3 LIMIT 10;
4
```

 Below the query editor, a green bar indicates '10 rows' of data. The data is displayed in a table with the following columns: holiday, temp, rain_1h, snow_1h, clouds_all, weather_main, weather_description, date_time, and traffic_volume. The table contains 10 rows of data, showing various weather conditions and traffic volumes.

holiday	temp	rain_1h	snow_1h	clouds_all	weather_main	weather_description	date_time	traffic_volume
	288.28	0	0	40	Clouds	scattered clouds	2012-10-02 09:00:00	5545
	289.36	0	0	75	Clouds	broken clouds	2012-10-02 10:00:00	4516
	289.58	0	0	90	Clouds	overcast clouds	2012-10-02 11:00:00	4767
	290.13	0	0	90	Clouds	overcast clouds	2012-10-02 12:00:00	5026
	291.14	0	0	75	Clouds	broken clouds	2012-10-02 13:00:00	4918
	291.72	0	0	1	Clear	sky is clear	2012-10-02 14:00:00	5181
	293.17	0	0	1	Clear	sky is clear	2012-10-02 15:00:00	5584
	293.86	0	0	1	Clear	sky is clear	2012-10-02 16:00:00	6015
	294.14	0	0	20	Clouds	few clouds	2012-10-02 17:00:00	5791
	293.1	0	0	20	Clouds	few clouds	2012-10-02 18:00:00	4770

1.2 Analyse annual traffic trends

Compare the **total yearly traffic volume** for:

- 2012
- 2013
- 2014
- 2015
- 2016
- 2017

Then:

- Identify years with increases or decreases.
- Calculate the change between years where appropriate.
- Explain at least **two meaningful observations** about traffic volume trends.

ExtendsClass SQLite browser online

Free Online Toolbox for developers

Database file▼ Queries▼ Save & Share Export▼ Run ►

Your database

Table
traffic

View

SQL Documentation

SELECT
UPDATE
DELETE
INSERT
CREATE TABLE

```

1 SELECT
2     strftime('%Y', date_time) AS year,
3     SUM(traffic_volume) AS total_traffic_volume
4 FROM traffic
5 WHERE strftime('%Y', date_time)
6     BETWEEN '2012' AND '2017'
7 GROUP BY strftime('%Y', date_time)
8 ORDER BY year;

```

6 rows

year	total_traffic_volume
2012	8208767
2013	28177412
2014	15731289
2015	14181206
2016	29494821
2017	35428156

Database file▼ Queries▼ Save & Share Export▼ Run ►

```

1 SELECT
2     strftime('%Y', date_time) AS year,
3     COUNT(*) AS records
4 FROM traffic
5 WHERE strftime('%Y', date_time) BETWEEN '2012' AND '2017'
6 GROUP BY year
7 ORDER BY year;

```

6 rows

year	records
2012	2559
2013	8573
2014	4839
2015	4373
2016	9306
2017	10605

Task 1.2 — Annual Traffic Trends

Year	Records	Total Traffic Volume	Change from Previous Year	Trend	(
2012	2,559	8,208,767	—	Baseline	
2013	8,573	28,177,412	+19,968,645	Increase	
2014	4,839	15,731,289	–12,446,123	Decrease	
2015	4,373	14,181,206	–1,550,083	Decrease	
2016	9,306	29,494,821	+15,313,615	Increase	
2017	10,605	35,428,156	+5,933,335	Increase	

Observation 1 — Traffic volume fluctuated considerably across the period.

Total recorded traffic volume increased from 8.21 million in 2012 to 28.18 million in 2013, before decreasing in both 2014 and 2015. It then increased strongly in 2016 and reached the highest recorded total of 35.43 million in 2017.

Observation 2 — Differences in annual totals should be interpreted carefully.

The number of records varies substantially between years. For example, 2012 contains 2,559 observations, compared with 8,573 in 2013, while 2014 and 2015 also contain fewer observations than 2016 and 2017. Therefore, some differences in total traffic volume may be caused by unequal data coverage rather than changes in actual traffic demand alone.

1.3 Analyse temperature around holidays

Compare temperature patterns for **2015, 2016 and 2017** during:

- New Year's Day
- Labor Day

Identify notable year-on-year changes and discuss whether these changes appear relevant to traffic conditions.

Database file▼

Queries▼

Save & Share

Export▼

Run ►

```
1 SELECT DISTINCT
2     strftime('%Y', date_time) AS year,
3     holiday,
4     temp,
5     traffic_volume
6 FROM traffic
7 WHERE holiday IN ('New Years Day', 'Labor Day')
8 AND strftime('%Y', date_time) BETWEEN '2015' AND '2017'
9 ORDER BY year, holiday;
```

5 rows

year	holiday	temp	traffic_volume
2015	Labor Day	295.02	973
2016	Labor Day	293.17	1064
2016	New Years Day	265.94	1513
2017	Labor Day	295.54	1026
2017	New Years Day	270.62	798

Task 1.3 — Temperature Around Labor Day and New Year's Day

Holiday	Year	Temperature (K)	Change vs Previous Available Year (K)	Traffic Volume 
Labor Day	2015	295.02	—	973
Labor Day	2016	293.17	−1.85	1,064
Labor Day	2017	295.54	+2.37	1,026
New Year's Day	2015	No data	—	—
New Year's Day	2016	265.94	—	1,513
New Year's Day	2017	270.62	+4.68	798

Observation 1 — Labor Day

Labor Day temperatures were relatively stable across the three available years. Temperature decreased from 295.02 K in 2015 to 293.17 K in 2016, a fall of 1.85 K, before increasing to 295.54 K in 2017, a rise of 2.37 K. Traffic volume changed only moderately, from 973 to 1,064 and then 1,026 vehicles. Based on these three observations, there is no clear indication that the relatively small temperature changes had a strong effect on Labor Day traffic.

Observation 2 — New Year's Day

There is no New Year's Day record for 2015 in the supplied dataset, so a complete three-year comparison isn't possible. Between the available years, temperature increased from 265.94 K in 2016 to 270.62 K in 2017, an increase of 4.68 K, while traffic volume decreased from 1,513 to 798 vehicles. Although the warmer year had lower traffic, two observations are insufficient to conclude that temperature caused the reduction. Other factors such as holiday travel behaviour, weather conditions, and incomplete data coverage may also influence traffic.